Half-amplitude confinement on one fixed-target probability panel

Bounded research note

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1 Scope and retained entry

Let an actual fixed-target maximizing law at $n \ge N = 42$ have nonnegative excess $\mathfrak{a}$ over $c = C_{\mathrm{E}}$. Retain fixed-target contact, the sharp predecessor $C_{n-1} \le c$, and the accepted four-pass signed entry cover, including its complete Fourier estimates and ordered same-law acquisition. The previous panel result supplies, on $p \in [l, h] = [0.41, 0.42]$,

$$\theta < 0.054, \quad d < 0.069, \quad \rho < 1.22, \quad 0.09736 \le \beta \le 0.234, \quad 1.9706 < \ell < 2.034, \quad \lambda < 0.6,$$

$$\operatorname{supp} Y \subset (-0.6, 1.6), \qquad |U| < 0.6, \quad X = \ell(Y - p), \quad Y = B + U, \quad \mathbb{E}(U \mid B) = 0.$$

Here $\theta = \mathbb{E}(X - \mathbb{E}[X \mid B])^2$, $\lambda = n\mathbb{E}U^2$, and $d = 1 - (A + B)/\ell$ uses the actual central gap $(-A, B)$. The physical defect is $D = (3 + \sqrt{10})\rho - \gamma - 3J$. The joined result below proves $|U| < 1/2$ on this panel. The analytic entry and Fourier premises remain explicit; no count comparison follows from this support statement alone.

2 A whole-rectangle resource from actual moment cells

For each retained moment cell, write its proved scalar raw bound as

$$\mathcal{R}_c \le K_F + cA + H(cK_1 + \varphi W) + s|H| - q(c)H^2, \qquad \varphi = \phi(H/\sqrt{n}). \tag{1}$$

The cell retains ranges for $\rho, \beta, \gamma, m, p$ and the signed quantities K_1, W, with the same-law density license $\varphi \in [\varphi_-, 2/5]$. Its separately proved raw ceiling is $T < Nc_-\rho_-$; put

$$\eta = (c_-\rho_- - T/N)/2 > 0.$$

The fixed-target phase-price identity then gives

$$\mathcal{U} := (n + H^2)\mathfrak{a} + n\phi(0)D/6 + n^{3/2}L_{\mathrm{leak}} \le \mathcal{R}_c - \eta H^2. \tag{2}$$

All quantities in these inequalities refer to one law and raw threshold. In particular, $\mathcal{U}$ is a joint resource, not just a defect bound.

For each endpoint $t \in \{c_-, c_+\}$, define

$$B_t = K_F + tA, \quad Q_t = q(t) + \eta,$$

$$F_{t,+} = \max(tK_1 + \varphi W) + s, \qquad F_{t,-} = -\min(tK_1 + \varphi W) + s,$$

where maxima and minima range over all eight corners of the recorded K_1, φ, W rectangle. Their multiaffine form makes the corner evaluation valid on the full rectangle. For fixed $|H|$, maximizing affine functions of c is convex, so the two endpoints bound every $c \in [c_-, c_+]$. Also $Q_t > 0$ is explicitly checked. Thus, for each sign ϵ of H,

$$\mathcal{U} \le \min\left\{T, \max_{t \in \{c_-, c_+\}} (B_t + F_{t,\epsilon}|H| - Q_t H^2)\right\}. \tag{3}$$

The first entry T is a valid, possibly loose, upper bound because $\mathcal{U} \le \mathcal{R}_c$.

Lemma 1 (Necessary noise filtering)**.** *Let $v = p(1-p) \in [v_-, v_+]$ and $0 \le \lambda \le \lambda_+$. Every actual law at every $n \ge N$ satisfies*

$$\theta \le \Theta_\lambda := \frac{\lambda_+}{Nv_- + \lambda_+}, \qquad \sqrt{\frac{v_-^2}{v_- + \lambda_+/N}} \le m \le \sqrt{v_+},$$

$$1 - 2h - 2(1-l)\Theta_\lambda \le \beta \le 1 - 2l + 2h\Theta_\lambda.$$

Proof. The exact identities are

$$\lambda = \frac{nv\theta}{1-\theta}, \qquad m^2 = v(1-\theta) = \frac{v^2}{v + \lambda/n}.$$

The last expression increases with v and n and decreases with λ. This proves both moment bounds, including the limit $n \longrightarrow \infty$ in the upper bound. Conditional centering gives $\beta = (1-\theta)(1-2p) + \mu_+(\mathbb{R}) - \mu_-(\mathbb{R})$, where the two nonnegative masses sum to θ. Hence

$$1 - 2p - 2(1-p)\theta \le \beta \le 1 - 2p + 2p\theta.$$

Replacing each coefficient outward on $p \in [l, h]$ proves the stated conservative bounds. The argument also covers zero noise and does not assume $p \le 1/2$. □

For a closed rectangle $|H| \in [h_0, h_1]$, $\lambda \in [\lambda_0, \lambda_1]$, discard a moment cell only if its p, m, β range is disjoint from the necessary ranges, or if its proved upper bound on $|H|$ is strictly below h_0. The noise filter uses only λ_1. For every remaining cell, clip h_1 by that cell's phase ceiling. Maximize each concave quadratic in (3) at the two endpoints and, when included, its vertex. Take the maximum over both target endpoints and every remaining cell, and the cellwise minimum with T. Call the result S_ϵ. This bounds $\mathcal{U}$ throughout the closed rectangle. All equalities at seams and all intersecting moment cells remain included. If no cell remains, the rectangle contains no law allowed by the ancestor cover. If $S_\epsilon < 0$, nonnegativity of $\mathcal{U}$ excludes the rectangle.

3 The retained confinement gates

Keep exactly the preceding $|U| < 0.6$ strip, target $r_0 = 1/2$, inward bound $d_0 = 0.069$, and both buffers $\tau = 1/5$. The geometric gates are

$$1 - 0.6 - 2d_0 > 0, \quad d_0 + 2\tau < r_0, \quad 1 - d_0 - \tau > 0, \quad 1 - 2d_0 - 2\tau > 0.$$

The lower intervals also have strict diameter below the actual increment. The deleted sums retain indices $n-1, n-2 \ge 40$ and the antichain cap; no second deleted-score hypothesis is substituted for that proof. The signed slopes, positivity gates, same-law moment coefficients and two resource error families are unchanged from the preceding test. On each rectangle replace only its phase resource by S_ϵ.

In particular the remaining moment resource is

$$R_* = S_\epsilon/\kappa - b_* E^2 \lambda_0, \qquad \kappa = 997/15000, \quad E = 1.9706.$$

A strictly negative value excludes the rectangle. Equality is not automatically excluded: the new bound is non-strict, and all subsequent inverse-index sign gates are retained. The old and tangent error bounds are each valid for the same resource simplex; their minimum is used. Every accepted contact contradiction is a strict positive margin.

4 The two unassisted diagnostics

Use eight equal closed phase bands of width $907/4000 = 0.22675$ up to 1.814, eight equal closed noise bands of width $3/40 = 0.075$ up to 0.6, and both signs. There are exactly 128 boxes. The new calculation excludes 87 boxes: 16 have no incident moment cell, 6 have negative joint resource, and 65 have insufficient resource for their lower noise endpoint. The other 41 boxes have 82 ray gates; 74 pass and 8 fail. Every lower-ray gate passes.

All failures are upper-ray gates on the first noise band $[0, 0.075]$. In the table, j denotes the closed phase band $[907j/4000, 907(j+1)/4000]$. Displayed margins are approximations only; the log records exact rationals.

sign of H	j	contact side minus error
$+$	1	-0.004052969
$+$	2	-0.015160061
$+$	3	-0.013432901
$-$	1	-0.008291001
$-$	2	-0.025939179
$-$	3	-0.035976266
$-$	4	-0.034114818
$-$	5	-0.001378038

The worst box has $H < 0$, $|H| \in [0.68025, 0.907]$. Its resource ceiling is approximately 0.6695213783, contact side 0.2349977631, and error majorant 0.2709740296. Failure of this sufficient bound is neither a counterexample to confinement nor evidence for a maximizing law in the box. The zero-noise case itself has $U = 0$ exactly; that does not remove the positive-noise portion of these closed boxes.

One predeclared redistribution keeps eight noise bands, with endpoints

$$0, \quad 0.01, \quad 0.025, \quad 0.05, \quad 0.075, \quad 0.15, \quad 0.30, \quad 0.45, \quad 0.60.$$

On this second 128-box cover, 55 boxes are excluded and 145 of the 146 ray gates pass. The only failure is the upper ray with $H < 0$, $|H| \in [0.68025, 0.907]$, $\lambda \in [0.05, 0.075]$; its margin is approximately -0.0075629738213. No further grid is introduced.

5 A direct all-threshold low-noise support proof

Lemma 2. *Under the retained panel hypotheses,* $0 \le \lambda \le 3/40$ *implies* $|U| < 1/2$ *for every raw threshold* H *and every actual* $n \ge 42$.

Proof. Put $v_- = l(1-l)$, $v_+ = h(1-h)$ and

$$\Theta_* = \frac{3/40}{42v_- + 3/40}, \qquad E_* = 2.0186650184 < \sqrt{(1-\Theta_*)/v_+}.$$

Lemma 1 gives $\theta \le \Theta_*$ and $\ell \ge E_*$. The tighter endpoint bounds, using monotonicity in p, are

$$b_- = 1 - 2h - 2(1-h)\Theta_* \le \beta \le 1 - 2l + 2l\Theta_* = b_+.$$

Both endpoint functions decrease with p because $\Theta_* < 1$.

The acquired $|U| < 0.6$, conditional centering, and sign preservation give the exact third-moment expansion

$$\rho = \ell^3\{v(1-2v) + 3\mathbb{E}|B-p|U^2 + \mathbb{E}[\operatorname{sgn}(B-p)U^3]\}.$$

Since $p < 1/2$, $|B - p| \le 1 - p \le 1 - l$, while $|U|^3 \le 0.6U^2$. The positive skeleton term is at most $\rho_B(l) = (1 - 2v_-)/\sqrt{v_-}$. Also $\ell^3 \le v_-^{-3/2}$, and the exact gate

$$24^2 v_-^3 > [3(1 - l) + 0.6]^2$$

therefore gives $\rho \le \rho_B(l) + 24\lambda/n$. With $s_- = 0.491833305 < \sqrt{v_-}$, use the rational upper

$$r_* = (1 - 2v_-)/s_- + 24(3/40)/42 < 1.092400.$$

All these are consequences of the same conditional marginal, with the previous strip used before the new one is acquired.

Take the exact rational boundaries

$$u_- = E_*(l + 1/2) = 1.836985166744, \qquad u_+ = E_*(1 + 1/2 - h) = 2.180158219872.$$

Both exceed $\sqrt{3}$. For $u > \sqrt{3}$, put

$$A_u = (u^3 - 3u)/6, \quad H_4 = u^4 - 6u^2 + 3, \quad S = 81/80,$$

$$E_u = u^5/4100 + 3u/820 + 17(3u^2 - 2)/78720, \quad Q_{r,b} = -u^3 + 3ru^2/2 + 3bu + (8/875)r.$$

The fourth-order heat expansion with absolute quartic coefficient bounds the support contact by

$$P_{r,b}(u) = \tfrac{2}{5} S^3 (A_u + |H_4|/72) + S^3 E_u + 0.4095 Q_{r,b}(u),$$

provided $Q_{r,b} < 0$. Its remainder is valid on this domain because $u^2 - 1 > 0$; it is uniform in every raw H. At $(u_-, r_*, -b_-)$ and (u_+, r_*, b_+) the exact checker obtains

$$P_{r_*,-b_-}(u_-) < -0.51645700, \qquad P_{r_*,b_+}(u_+) < -0.24695345.$$

All twelve continuation signs are strictly negative: on each ray check Q, Q', Q'' and L, L', L'', where $L = (2/5)(u^2/2 + 1/2 + 8/875) + 0.4095Q'$. The higher derivatives $Q''' = -6$ and $L'' = 2/5 - 6(0.4095) < 0$ preserve the signs. The all-threshold derivative heat bound then makes the contact upper bound decrease along the entire exterior ray. Thus support, including atoms at the boundaries, lies in $(-u_-, u_+)$.

Since $\ell \ge E_*$ and $p \in [l, h]$, this implies $Y = X/\ell + p \in (-1/2, 3/2)$ strictly. The already proved inward bound gives $U_0 \le d < 0.069$ and $U_1 \ge -d > -0.069$, so both conditional supports satisfy $|U| < 1/2$. At zero noise conditional centering instead gives $U_0 = U_1 = 0$ directly. No division by noise is used. □

Proposition 1 (Complete conditional panel confinement)**.** *On the retained panel, every actual $n \ge 42$ satisfies $|U| < 1/2$. The inward bound $d < 0.069 < 1/4$ remains valid.*

Proof. Use the original equal 128-box cover. Lemma 2 covers all 16 phase/sign boxes on the closed lowest-noise band $[0, 0.075]$. Among the other 112 boxes, 84 are excluded by the validated incident-cell or resource conditions, and 28 pass both strict ray inequalities. The smallest of those 56 strict margins exceeds 0.00892011917. Every noise and phase seam is included. In particular $\lambda = 0.075$ is covered by the direct lemma and by the adjoining closed high-noise band. All high-noise phases lie in the acquired $|H| < 1.814$ interval; the low-noise lemma requires no phase restriction. □

6 Reproduction and acceptance boundary

The final computation reads frozen source copies and a frozen phase export, verifies their byte pins, and proves the export instrumentation reversible to the original signed kernel. It freshly reproduces the entire last signed stage: 2048 original moment boxes, 682 active cells. It compares the full ordered active-cell output, not a stored success marker, and reconstructs all 1364 signed endpoint quadratics before joining resources. Pinned local modules are compiled directly from their verified source bytes. The frozen producer root must contain exactly the listed files and no symlinks; this excludes an extra module shadowing a later search path. The child producer uses a new empty bytecode-cache prefix together with `-B`; the prefix is checked to remain empty. Source pins alone or `-B` alone would not exclude execution of an existing bytecode cache. Ten malformed or false controls reject, including missing, duplicate, reordered and sign-swapped cells, altered curvature, an unlicensed phase price, and a false unassisted whole-panel success claim. The direct heat lemma has five separate adverse controls.

This fresh execution does not re-prove the accepted earlier entry stages, the analytic Fourier estimates, or the contact theorem. Mathematical status: the resource join and direct lemma prove the stated conditional all-index panel confinement. Execution status: fresh exact replay and composed acceptance, with both earlier diagnostics preserved. No native certificate campaign, finite-base acceptance, count comparison, or global theorem is claimed.